

🏛 Academic Employment

Aalto University

Postdoctoral Researcher, Advisor Dr. Samuel Kaski

June 2024–Present

Espoo, Finland

🎓 Education

Purdue University

PhD Statistics, Advisor: Dr. Anindya Bhadra

2019 – 2024

Indiana, USA

Thesis title: *On the use of α -stable random variables in Bayesian bridge regression, neural networks and kernel processes*

Universidad de Costa Rica

BS Pure Mathematics

2015 – 2019

San José, Costa Rica

Universidad de Costa Rica

BS Actuarial Science

2014 – 2017

San José, Costa Rica

🔍 Research Interests

- Heavy tails
- High dimensional data
- Bayesian inference
- Gaussian processes
- Computational methods
- Causal inference

📖 Publications

📄 Preprints

» Fernández-Loría, C. and **Loría, J.** (2026+), Inferring Effect Ordering Without Causal Effect Estimation (*submitted*) [\[arxiv\]](#)

📄 Publications

- » Mäkinen L., **Loría, J.**, and Kaski, S. (2026+), Bayesian Meta-Learning with Expert Feedback for Task-Shift Adaptation through Causal Embeddings (*accepted to ICML 2026*) [\[arxiv\]](#)[\[github\]](#)
- » Björkman, Z., **Loría, J.**, Wharrie, S., and Kaski, S. (2026), Incorporating Expert Knowledge into Bayesian Causal Discovery of Mixtures of Directed Acyclic Graphs (*accepted to AISTATS 2026*) [\[arxiv\]](#)[\[github\]](#)
- » **Loría, J.** and Bhadra, A. (2025), Deep Kernel Posterior Learning under Infinite Variance Prior Weights. *The 13th International Conference on Learning Representations (ICLR 2025)*. [\[link\]](#) [\[github\]](#)
- » **Loría, J.** and Bhadra A. (2024). Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance. *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, **PMLR 244** 2331-2349 [\[link\]](#) [\[github\]](#)
- » **Loría, J.** and Bhadra, A. (2024). SURE-tuned bridge regression. *Statistics and Computing* **34**, 30. [\[doi link\]](#) [\[github\]](#)
- » Viquez, J. J., Viquez, J., Campos, A., **Loría, J.**, Mendoza, L. A. (2018). Demographic Modeling via 3-dimensional Markov Chains *Revista de matemática Teoría y aplicaciones* available in <https://revistas.ucr.ac.cr/index.php/matematica/article/view/33608>

🏆 Awards

Somos Purdue Award

Class of 2024

recognition of commitment to excellence and for leading with *ganas* at Purdue University.

Bravo Award

Spring 2024

one time staff award

Ross-Lynn Research Scholar Grant

Spring 2024

provides funding for notable graduate students

Fully funded assistance to workshop on Bayesian Statistics and Statistical Learning

December 2023

IMSI, Illinois, USA

Department of Statistics Travel Award	Summer 2023
Department of Statistics Teaching Academy Associate Nominee	Spring 2023
Emily and Paul Kidwell graduate student excellence Award	Spring 2022
given to graduate students in the Department of Statistics in Purdue University interested in careers in academia.	
Bob and Marjorie McLean Scholarship	Summers 2020 and 2021
given to distinguished graduate students of the Department of Statistics in Purdue University.	
Bronze Medal	2015
awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas	
Honorable Mention	2014
awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas	

Talks and Posters Presented

Deep Kernel Posterior Learning under Infinite Variance Prior Weights	April, 2025
<i>Poster in International Conference on Learning Representations</i>	Singapore
Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance	July, 2024
<i>Poster in Uncertainty in Artificial Intelligence</i>	Barcelona, Spain
Predicción en redes neuronales Bayesianas infinitas con pesos de varianza infinita	May 22nd, 2024
<i>Seminario Intersede</i>	CIMAT, Guanajuato, México
SURE-tuned Bridge Regression	August 7th, 2023
<i>Contributed Paper in Joint Statistical Meeting</i>	Toronto, Canada
SURE-tuned Bridge Regression	July 10th, 2023
<i>Contributed talk in XVI Latin American Congress of Probability and Mathematical Statistics</i>	São Paulo, Brazil
SURE-tuned Bridge Regression	May 26th, 2023
<i>Poster in Statistics and Optimization in Data Science Workshop, in Purdue Business School</i>	Indiana, USA
SURE-tuned Bridge Regression	April 2023
<i>Invited talk in a weekly colloquium at CIMPA Universidad de Costa Rica.</i>	Online
Learner Culture and Perceptions of Social Media Applications and Learning Management Systems for Language Learning	April 2023
<i>Paper session American Educational Research Association (AERA)</i>	Illinois, USA
Examining Learner Culture, Country, Gender, and English Public Speaking and Peer Feedback Self-Efficacy	April 2023
<i>Round table in AERA</i>	Illinois, USA
Learning the Ranking of Causal Effects with Confounded Data	October 2022
<i>Flash talk in INFORMS</i>	Indiana, USA

Research Experience

Purdue University

- **Research Assistant to Dr Anindya Bhadra** Spring 2022 and Fall 2023
 - » in the project Inference in Infinite Width Neural Networks with Infinite Variance
- **Research Assistant to Dr Arman Sabbaghi** Fall 2020-Summer 2021
 - » designing experiments and evaluating results for refinement of nanotellurene thickness
- **Team leader in Statistics for the Community** 2019-2021
 - » investigated the effects of having a bicycle to enable access to charity locations for impoverished people, with the LifeCycle program (*voluntary*).

Universidad de Costa Rica

- Research Assistant to Dr Luis Barboza

2016

» implementing an actuarial model for a projection model of Caja Costarricense del Seguro Social

- Research Assistant to Dr Juan José Víquez

2015

» implementing a multivariate Markov chains model for the projection of public universities in Costa Rica

👥 Service to the Profession

PhD. Thesis advisor

» Lotta Mäkinen (ongoing, with S. Kaski as supervisor)

MSc. Thesis advisor

» Zachris Björkman (2024), Lotta Mäkinen (2025), and Axel Hedman (2025)

Reviewer

» Statistics in Medicine (2023), Neurips (2023), UAI (2024, 2026), ICLR (2024, 2026), AISTATS (2026), ICML (Silver Reviewer: 2026)

✍ Teaching Experience

Aalto University

Instructor

Fall 2022

Espoo, Finland

» Co-organized a seminar on causal inference, following Pearl's 'primer' book on causal inference.

Purdue University

Instructor

Fall 2022

Indiana, United States

» Taught one section of the course *Introduction to Statistics*. An introductory course for undergraduate students of computer science, mathematics, actuarial science, and statistics majors. Responsibilities included three weekly lectures, preparation of the relevant material, grading, and collaboration with the other instructors in the evaluation and design of the assignments of the course.

Purdue University

Instructor

Fall 2021 – Summer 2022, Summer 2023

Indiana, United States

» Taught two sections of the course *Elementary Statistical Methods*. An undergraduate level course for non-stem majors. Responsibilities included two weekly lectures to each section, preparation of the material, grading, leading teaching assistants and collaboration with other instructors for the evaluation of the course assignments.

Purdue University

Teaching Assistant

Fall 2019 – Spring 2021

Indiana, United States

» Held laboratories for two sections, office hours, and grading for the course *Elementary Statistical Methods*.

Universidad de Costa Rica

Instructor

Fall of 2017 & 2018

San José, Costa Rica

» Taught undergraduate course on basic functionalities of Excel and R. Responsibilities included lesson planning, course scheduling, and preparation of homework, exams, and course projects.

👜 Industry Experience

Banco Promérica

Financial Risk Analyst, analyzed financial data to evaluate and forecast credit risk

January 2019 – July 2019

San José Costa Rica

Quanto Soluciones S.A.

Actuarial Model Programmer, implemented credit risk models

January 2017 – July 2018

San José Costa Rica

Jorge Loría

PhD in Statistics

 Jorge Loría  /loriaJ  jorge.loria@aalto.fi
 /jorge-loria  loriaj.github.io/

Communication

Spanish *Native*

English *Advanced*

Portuguese *Basic*

Leadership Roles

- **President of Statistics Graduate Student Organization**

2021-2022
Indiana, USA

- **Vice-president of Costa Rican Students Association at Purdue (CRISaP)**

2021-2024
Indiana, USA